

Pirate Balancing Scale Puzzle

Logic Map & System Design

Puzzle Overview

Players must balance five different spice pouches using gold coins of various denominations. The scale appears to be a traditional balance scale from the 1600s pirate era, but uses RFID technology to detect correct combinations. A compass indicator provides period-authentic feedback.

Physical Components

Component	Description	Technology
Balance Scale	Two-plate traditional scale (theatrical)	Display prop
Spice Pouches	5 pouches with different weights	RFID tags embedded
Gold Coins	3 denominations with weights printed	RFID tags embedded
Compass Indicator	Points to "Aye" or "Nay"	Servo motor controlled
Left RFID Reader	Detects spice pouch	RC522 or PN532
Right RFID Reader	Detects coin combination	RC522 or PN532
ESP32 Controller	Processes logic and controls servo	ESP32-S3

Weight Configuration

Spice Pouch	Weight	RFID Tag	Required Coins
Pouch 1	1.5g	Tag 112	1g + 0.5g or 0.75g + 0.75g
Pouch 2	1.75g	Tag 113	1g + 0.75g or 0.5g + 0.5g + 0.75g
Pouch 3	2.25g	Tag 114	1g + 1g + 0.25g or 0.75g + 0.75g + 0.75g
Pouch 4	3.5g	Tag 115	1g + 1g + 1g + 0.5g or various combos
Pouch 5	4.0g	Tag 116	1g + 1g + 1g + 1g

Coin Type	Weight	RFID Tag Range
Gold Doubloon	1.0g	Tags 200-210

Half Doubloon	0.5g	Tags 211-220
Piece of Eight	0.75g	Tags 221-230
Quarter Doubloon	0.25g	Tags 231-240

System Logic Flow

1. System continuously monitors left RFID reader for spice pouch detection
2. When spice pouch is detected, system retrieves the target weight from lookup table
3. System continuously monitors right RFID reader for coin tags
4. System calculates total weight of detected coins based on their RFID tags
5. System compares calculated coin weight to required spice pouch weight
6. If weights match: Servo rotates compass from 'Nay' to 'Aye' position
7. If weights don't match: Compass remains at 'Nay' position
8. When spice pouch is removed from left reader: Servo resets compass to 'Nay' position
9. System returns to step 1 (ready for next attempt)

System States

State	Condition	Compass Position	Action
IDLE	No spice pouch detected	Nay	Monitor left reader
DETECTING	Spice pouch on left	Nay	Monitor right reader & calculate
BALANCED	Correct coin combo	Aye	Hold until pouch removed
RESET	Pouch removed	Nay	Return to IDLE

Hardware Connections

ESP32-S3 Pin Configuration:

Component	ESP32 Pin	Notes
Left RFID SDA	GPIO 21	I2C or SPI depending on reader
Left RFID SCL	GPIO 22	I2C or SPI clock
Right RFID SDA	GPIO 18	Second I2C/SPI bus or different CS pin
Right RFID SCL	GPIO 19	Second I2C/SPI clock
Servo Signal	GPIO 25	PWM capable pin
Power (5V)	5V	For RFID readers and servo

Ground	GND	Common ground for all components
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Programming Logic Overview

Data Structures:

Spice Pouch Lookup Table: Maps RFID tag IDs to weight values in grams

Coin Weight Lookup Table: Maps RFID tag ID ranges to coin weights

State Variables: Current spice pouch detected, calculated coin weight, compass position

Main Program Loop:

- Read left RFID reader for spice pouch tag
- If spice pouch detected:
 - Look up required weight
 - Read all tags from right RFID reader
 - Calculate total weight of coins
 - Compare weights
 - Update compass position via servo
- If no spice pouch detected:
 - Reset compass to 'Nay' position
- Delay 100-200ms before next loop iteration

Compass Servo Control

Position	Angle	PWM Value	Meaning
Nay	0°	~1000µs pulse	Incorrect or no attempt
Aye	90°	~1500µs pulse	Correct balance achieved

Player Experience Flow

1. Players discover five spice pouches and various gold coins with weights printed on them
2. Players place a spice pouch on the left side of the balance scale
3. Players experiment with different coin combinations on the right side

4. When correct total weight is achieved, compass dramatically spins to 'Aye'
5. Players note the successful combination and remove the spice pouch
6. Compass resets to 'Nay', ready for the next spice pouch
7. Players repeat process for all five spice pouches
8. Once all five combinations are discovered, the puzzle is complete

Implementation Notes

- RFID readers should be concealed beneath or within the scale plates
- Servo motor should be hidden behind compass display
- Consider adding sound effects (creaking wood, wind) when compass moves
- Test RFID read range to ensure reliable detection through scale materials
- Multiple coins of same denomination should be supported (same weight, different tags)
- Consider adding a 'tolerance' margin if using actual weight sensors as backup
- Compass movement should be smooth and deliberate (not jerky) for theatrical effect
- System should debounce RFID readings to prevent flickering between states